

Carry-Over Lottery Allocation: Practical Incentive-Compatible Drafts

Paper Track: Basketball

Open-source repository: <https://github.com/kvr06-ai/cola-sloan-ssac27>

Introduction

How can a league help weak teams through the draft without encouraging teams to tank? Carry-Over Lottery Allocation (COLA), presented here, is a family of draft mechanisms that all simultaneously discourage tanking and help the weakest teams.

Under any mechanism in the COLA framework, each team has a draft priority that rises and falls over time, but it does not rise based on more losses. That eliminates the tanking incentive. Draft priority carries over from year to year, being reduced when a team has playoff success or makes a high draft pick. That directs the most help to the weakest teams over time.

For 40 years, the NBA has navigated the tradeoff between helping weak teams and discouraging tanking. COLA solves that primary dilemma, but there are other tradeoffs worth considering among the COLA variants. After demonstrating COLA's incentive compatibility, this paper discusses the tradeoffs between mechanisms within the framework.

Methods

We identify several tunable dials for mechanisms within the COLA framework:

- Which teams are eligible for the lottery?
- How does a team's lottery priority increase?
- How does the priority decrease?
- How does the priority determine lottery odds and draft positions?

We consider different COLA variants plus other draft mechanisms, including 3-2-1 Lottery, and compare them using ZenGM Basketball-GM, an open-source simulator that plays out every game and roster move. We measure the mechanisms against these criteria:

- **Targeted help.** Does help reach the weakest teams, as measured by the difference between the help received by the highest priority team and other teams?
- **Long-term parity.** Can all teams be competitive, as measured by the longest drought any team has between playoff series wins?
- **Simplicity.** How complicated is the mechanism? Though measured subjectively, it is not difficult to identify complexity.
- **Robustness.** Does the mechanism admit tanking incentives at the margin? COLA variants are incentive-compatible under reasonable assumptions. Further relaxation of those assumptions may entail a choice between extra complexity or allowing potential tanking incentives in rare cases.

Results

Many mechanisms are dominated under these criteria, meaning there is another mechanism that is better on all four.

We ran every mechanism on the same 30 simulated leagues for 25 seasons each, 6,750 seasons in all. Because the leagues are shared, we compare mechanisms by paired differences with 95% confidence intervals. Simple COLA helps long-drought teams more than the current NBA lottery does, and more than 3-2-1 does. Both gains clear their confidence intervals. On parity, Simple COLA does at least as well as both. It never reads a team's record, so it offers no reward for losing. Simple COLA or Beckett COLA matches or beats every other mechanism in every column of Table 1. Parity is where every mechanism fell short. In every league some team went close to 25 seasons without winning a playoff series. No mechanism cut the number of never-winners by more than chance could explain.

Mechanism	Help to long-drought teams <i>draft places per drought season</i>	Reward for losing <i>draft places gained by finishing last</i>	Teams never winning a series <i>in 25 years, per league</i>	Longest drought <i>seasons, of 25</i>
Current NBA lottery (2019 to 2026)	0.07	1.32	0.63	23.3
New NBA lottery, 3-2-1 (from 2027)	0.02	losing costs picks	1.20	24.4
Equal-odds lottery	0.00	0	0.93	23.7
Classic COLA, the original version	0.10	1.85	0.93	24.1
Waitlist COLA	0.13	0	1.10	24.4
Simple COLA	0.24	0	0.60	23.3
Full-lottery COLA	0.09	0	0.83	23.8
Countdown COLA	0.19	losing costs picks	1.33	24.6
Beckett COLA	0.28	losing costs picks	0.83	23.7

Table 1. Nine mechanisms on the same 30 simulated leagues of 25 seasons each.

Conclusion

We identify a frontier of mechanisms that are not dominated and therefore viable options for implementation, depending on the league's goals.